

DBMS –PAYMENT SYSTEM MANAGEMENT

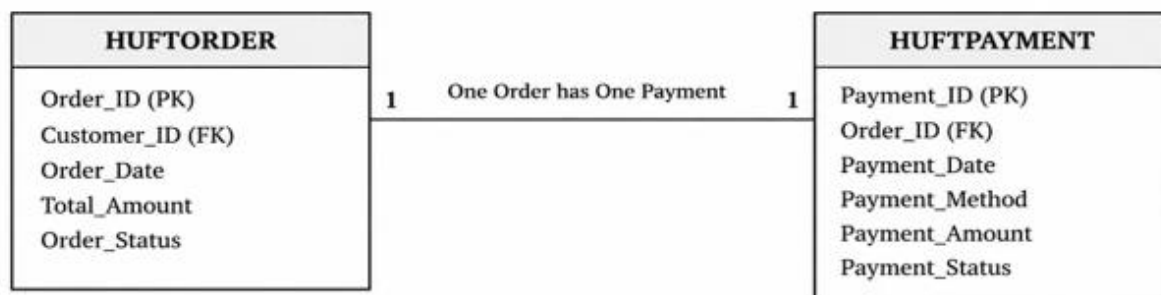
HUFT – E-Commerce Order Management Database System

1. DATABASE DESIGN

1.1 PAYMENT TABLE

Attribute	Description
Payment_ID	Unique payment ID
Order_ID	Related order ID
Payment_Date	Date of payment
Payment_Method	UPI / Card / Cash on Delivery / Net Banking
Payment_Amount	Amount paid
Payment_Status	Paid / Pending / Failed

2.TABLE RELATIONSHIP



3.TABLE CREATION

3.1 PAYMENT TABLE

```
SQL> CREATE TABLE HuftPayment (  
2     Payment_ID NUMBER PRIMARY KEY,  
3     Order_ID NUMBER REFERENCES HuftOrder(Order_ID),  
4     Payment_Date DATE NOT NULL,  
5     Payment_Method VARCHAR2(30),  
6     Payment_Amount NUMBER(10,2),  
7     Payment_Status VARCHAR2(20)  
8 );
```

Table created.

```
SQL> DESC HuftPayment;
```

Name	Null?	Type
PAYMENT_ID	NOT NULL	NUMBER
ORDER_ID		NUMBER
PAYMENT_DATE	NOT NULL	DATE
PAYMENT_METHOD		VARCHAR2(30)
PAYMENT_AMOUNT		NUMBER(10,2)
PAYMENT_STATUS		VARCHAR2(20)

3.2 VALUE INSERTION

```
SQL> INSERT INTO HuftPayment VALUES  
2 (1, 1001, TO_DATE('18-09-2026','DD-MM-YYYY'), 'UPI', 1800.00, 'Paid');
```

1 row created.

```
SQL> INSERT INTO HuftPayment VALUES  
2 (2, 1002, TO_DATE('22-09-2026','DD-MM-YYYY'), 'Credit Card', 1849.00, 'Paid');
```

1 row created.

```
SQL>  
SQL> INSERT INTO HuftPayment VALUES  
2 (3, 1003, TO_DATE('19-09-2026','DD-MM-YYYY'), 'Debit Card', 1449.00, 'Paid');
```

1 row created.

```
SQL>  
SQL> INSERT INTO HuftPayment VALUES  
2 (4, 1004, TO_DATE('19-09-2026','DD-MM-YYYY'), 'UPI', 1273.00, 'Paid');
```

1 row created.

```
SQL>  
SQL> INSERT INTO HuftPayment VALUES  
2 (5, 1005, TO_DATE('20-09-2026','DD-MM-YYYY'), 'Cash on Delivery', 1075.00, 'Pending');
```

1 row created.

```
SQL>  
SQL> INSERT INTO HuftPayment VALUES  
2 (6, 1006, TO_DATE('20-09-2026','DD-MM-YYYY'), 'UPI', 650.00, 'Paid');
```

1 row created.

```

SQL>
SQL> INSERT INTO HuftPayment VALUES
  2 (7, 1007, TO_DATE('20-09-2026','DD-MM-YYYY'), 'Net Banking', 1197.00, 'Paid');

1 row created.

SQL>
SQL> INSERT INTO HuftPayment VALUES
  2 (8, 1008, TO_DATE('21-09-2026','DD-MM-YYYY'), 'Credit Card', 1498.00, 'Paid');

1 row created.

SQL>
SQL> INSERT INTO HuftPayment VALUES
  2 (9, 1009, TO_DATE('21-09-2026','DD-MM-YYYY'), 'UPI', 998.00, 'Paid');

1 row created.

SQL>
SQL> INSERT INTO HuftPayment VALUES
  2 (10, 1010, TO_DATE('21-09-2026','DD-MM-YYYY'), 'Cash on Delivery', 900.00, 'Pending');

1 row created.

```

4. DISPLAY SUCCESSFUL PAYMENTS

```

SQL> SELECT *
  2 FROM HuftPayment
  3 WHERE Payment_Status = 'Paid';

```

PAYMENT_ID	ORDER_ID	PAYMENT_D	PAYMENT_METHOD	PAYMENT_AMOUNT
1	1001	18-SEP-26	UPI	1800
Paid				
2	1002	22-SEP-26	Credit Card	1849
Paid				
3	1003	19-SEP-26	Debit Card	1449
Paid				

PAYMENT_ID	ORDER_ID	PAYMENT_D	PAYMENT_METHOD	PAYMENT_AMOUNT
4	1004	19-SEP-26	UPI	1273
Paid				
6	1006	20-SEP-26	UPI	650
Paid				
7	1007	20-SEP-26	Net Banking	1197
Paid				

```

PAYMENT_ID  ORDER_ID PAYMENT_D PAYMENT_METHOD          PAYMENT_AMOUNT
-----
PAYMENT_STATUS
-----
          8      1008 21-SEP-26 Credit Card          1498
Paid

          9      1009 21-SEP-26 UPI              998
Paid

8 rows selected.

```

5. DISPLAY FAILED/PENDING PAYMENTS

```

SQL> SELECT *
      2 FROM HuftPayment
      3 WHERE Payment_Status = 'Pending';

PAYMENT_ID  ORDER_ID PAYMENT_D PAYMENT_METHOD          PAYMENT_AMOUNT
-----
PAYMENT_STATUS
-----
          5      1005 20-SEP-26 Cash on Delivery          1075
Pending

         10      1010 21-SEP-26 Cash on Delivery           900
Pending

```

6. MODIFY PAYMENT STATUS

```

SQL> UPDATE HuftPayment
      2 SET Payment_Status = 'Paid'
      3 WHERE Payment_ID = 5;

1 row updated.

SQL> SELECT *
      2 FROM HuftPayment
      3 WHERE Payment_ID = 5;

PAYMENT_ID  ORDER_ID PAYMENT_D PAYMENT_METHOD          PAYMENT_AMOUNT
-----
PAYMENT_STATUS
-----
          5      1005 20-SEP-26 Cash on Delivery          1075
Paid

```

7. ANALYZE PAYMENT METHODS

7.1 COUNT TRANSACTIONS BY PAYMENT METHOD

```
SQL> SELECT
  2     Payment_Method,
  3     COUNT(*) AS Total_Transactions
  4 FROM HuftPayment
  5 GROUP BY Payment_Method
  6 ORDER BY Payment_Method;
```

PAYMENT_METHOD	TOTAL_TRANSACTIONS
Cash on Delivery	2
Credit Card	2
Debit Card	1
Net Banking	1
UPI	4

7.2 TOTAL AMOUNT BY PAYMENT METHOD

```
SQL> SELECT
  2     Payment_Method,
  3     SUM(Payment_Amount) AS Total_Amount
  4 FROM HuftPayment
  5 WHERE Payment_Status = 'Paid'
  6 GROUP BY Payment_Method
  7 ORDER BY Payment_Method;
```

PAYMENT_METHOD	TOTAL_AMOUNT
Cash on Delivery	1075
Credit Card	3347
Debit Card	1449
Net Banking	1197
UPI	4721

8. PAYMENT TRANSACTION REPORT

```
SQL> SELECT
  2     p.Payment_ID,
  3     o.Order_ID,
  4     c.CID AS Customer_ID,
  5     c.CNAME AS Customer_Name,
  6     p.Payment_Method,
  7     p.Payment_Date,
  8     p.Payment_Amount,
  9     p.Payment_Status
 10 FROM HuftPayment p
 11 JOIN HuftOrder o
 12     ON p.Order_ID = o.Order_ID
 13 JOIN CUST3 c
 14     ON o.Customer_ID = c.CID
 15 ORDER BY p.Payment_Date DESC;
```

PAYMENT_ID	ORDER_ID	CUSTOMER_ID	CUSTOMER_NAME

PAYMENT_METHOD		PAYMENT_D	PAYMENT_AMOUNT PAYMENT_STATUS

2	1002	2 Anu	
Credit Card		22-SEP-26	1849 Paid
8	1008	8 Meena	
Credit Card		21-SEP-26	1498 Paid
10	1010	10 Nisha	
Cash on Delivery		21-SEP-26	900 Pending

PAYMENT_ID	ORDER_ID	CUSTOMER_ID	CUSTOMER_NAME
5	1005	5 Priya	
Cash on Delivery		20-SEP-26	1075 Paid
7	1007	7 Arun	
Net Banking		20-SEP-26	1197 Paid
6	1006	6 Divya	
UPI		20-SEP-26	650 Paid

PAYMENT_ID	ORDER_ID	CUSTOMER_ID	CUSTOMER_NAME
3	1003	3 Ravi	
Debit Card		19-SEP-26	1449 Paid
4	1004	4 Karthik	
UPI		19-SEP-26	1273 Paid
1	1001	1 Abi	
UPI		18-SEP-26	1800 Paid

9 rows selected.